

Set-1

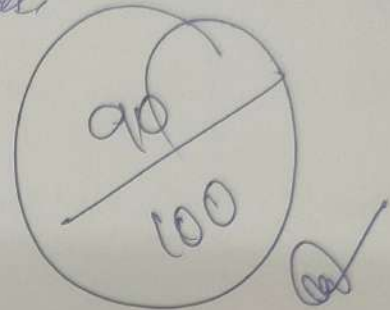
D) DES problem - one round of DES

DES normally uses a 64-bit plaintext block and a 64-bit key. For simplified one round calculation, suppose

Left half $L_0 = 12345678$

Right half $R_0 = ABCDEF01$

Round key $K_1 = 1A2B3C4D$



Expansion:

R_0 from 32 bits to 48 bits

$$E(R_0)$$

$$E(R_0) \oplus K_1$$

$$S(E(R_0) \oplus K_1)$$

Apply DES P-box

$$F(R_0, K_1) = P(S(E(R_0) \oplus K_1))$$

$$R_1 = L_0 \oplus F(R_0, K_1)$$

$$L_1 = R_0$$

$$R_1 = L_0 \oplus F(R_0, K_1)$$

2) Symmetric vs Asymmetric Encryption.

Feature	Symmetric Encryption	Asymmetric Encryption
Keys	one shared key	public key + private key
Speed	very fast	relatively slower
main purpose	Encrypt large amount of data	key exchange, authentication, digital signatures.
key distribution	Difficult because the secret key must be shared	Easier, because the public key can be distributed.
Examples	AES, ChaCha 20	RSA, ECC
Real time use	WhatsApp message / session encryption	Email encryption, certificate secure key exchange

WhatsApp:

WhatsApp uses end-to-end encryption primarily based on symmetric encryption for efficiently encryption message content.

Email:

Email system can use asymmetric cryptography such as PGP/S/MIME. A sender can information using the recipient's public key, and the recipient decrypts their private key.

3) RSA problem

Given

$$p = 7$$

$$q = 11$$

$$m = 9$$

$$n = p \times q$$

$$n = 7 \times 11 = 77$$

$$\phi(n) = (p-1)(q-1)$$

$$\phi(n) = 6 \times 10 = 60$$

$$1 < e < 60$$

$$\gcd(e, 60) = 1$$

$$e = 7$$

$$\gcd(7, 60) = 1$$

$$e \times d \equiv 1 \pmod{60}$$

$$7d \equiv 1 \pmod{60}$$

$$d = 43$$

$$7 \times 43 = 301$$

$$301 \pmod{60} = 1$$

RSA Key

Public

$$(e, n) = (7, 77)$$

Private

$$(d, n) = (43, 77)$$

$$c = m^e \pmod{n}$$

$$c = 9^7 \pmod{77}$$

$$9^2 = 81 \equiv 4 \pmod{77}$$

$$9^4 = 4^2 = 16$$

$$9^7 = 9^4 \times 9^2 \times 9$$

$$= 16 \times 4 \times 9$$

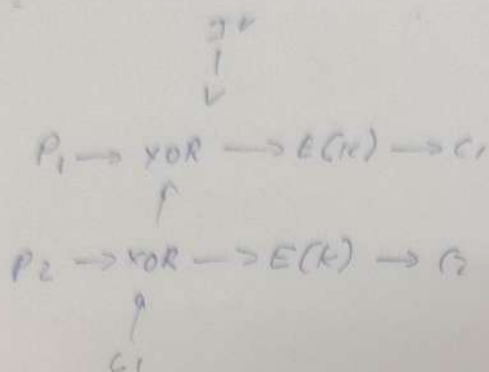
$$= 576$$

$$576 \pmod{77} = 37$$

$$\text{cipher } c = 37$$

4) CBC and CTR modes.

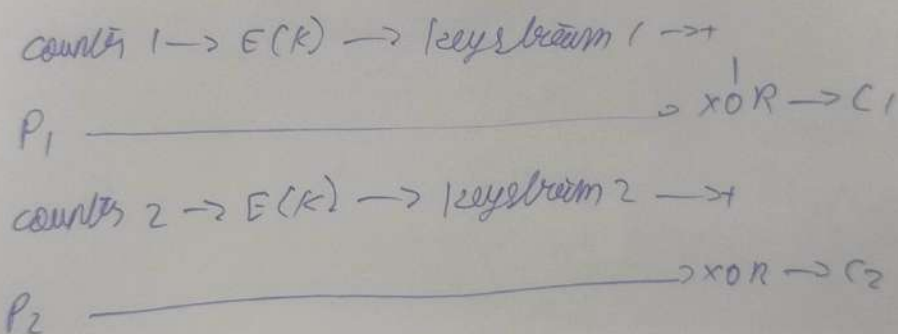
CBC:



$$C_1 = E(K, P_1 \oplus IV)$$

$$C_2 = E(K, P_2 \oplus C_1)$$

CTR



$$C_1 = P_1 \oplus E(K, \text{counter}_1)$$

$$C_2 = P_2 \oplus E(K, \text{counter}_2)$$

Each counter can be processed independently

- CBC encryption depends on the previous cipher text block
- CTR blocks are independent
- multiple counter values can be encrypted simultaneously.
- CTR also allows efficient random access to encrypted data.

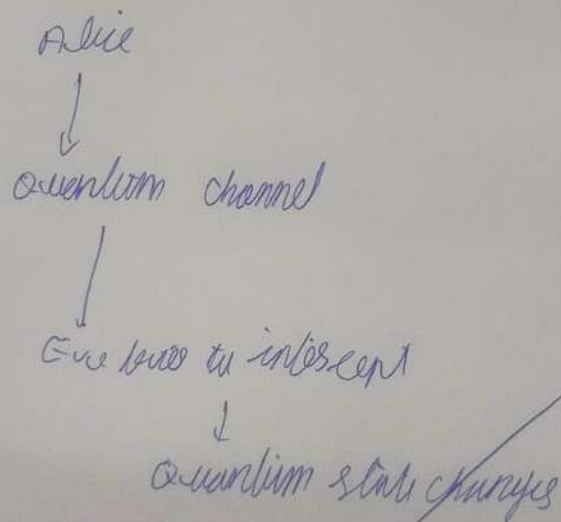
5) Quantum Cryptography and QKD

Quantum Cryptography:-

is a cryptographic approach that uses principles that of quantum mechanics to provide secure communication. The most important application is Quantum Key Distribution.

QKD distribution :-

In traditional cryptography, Alice and Bob need to securely obtain a shared secret key. If an attacker intercepts the key during transmission, the communication may be compromised.



main advantage

QKD does not rely only on the computational difficulty of mathematical problems. Its security is based on physical principles of quantum mechanics, making eavesdropping detectable.